

STUDENT PERFORMANCE PREDICTION USING MACHINE LEARNING

PROJECT REPORT

Course Name:

Artificial Intelligence

Submitted By:

Group Member 1: Abdullah — 4426-FOC/BSSE/F22

Group Member 2: Muhammad Khursheed — 4431-FOC/BSSE/F22

Group Member 3: Muhammad Sufyan Tipu — 4436-FOC/BSSE/F22

Group Member 3: AbuBakar Masood — 4564-FOC/BSSE/F22

Submitted To:

Sir Shakeel Ahmad

Institution:

International Islamic University, Islamabad

Department of Software Engineering

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Abstract

Student academic performance is a critical metric that directly influences educational policy, institutional planning, and individualized student support strategies. The ability to identify and predict at-risk students early in their academic journey enables educators and administrators to implement timely, targeted interventions. This study presents a machine learning-based approach to predict student academic performance using demographic and study-related features from the Students Performance in Exams dataset (Kaggle). Three classification algorithms were employed: Logistic Regression, Random Forest Classifier, and Support Vector Machine (SVM). The target variable was constructed as a binary classification label derived from the average score across mathematics, reading, and writing assessments, where a score of 60 or above was categorized as high performance (1) and below 60 as low performance (0). Comprehensive exploratory data analysis was conducted to examine score distributions, gender-based performance disparities, and the influence of parental education levels. The dataset was preprocessed through one-hot encoding, standard scaling, and an 80/20 train-test split. Evaluation metrics including accuracy, precision, recall, F1-score, and ROC-AUC were utilized to compare model performance. Results demonstrated that all three models achieved competitive predictive accuracy, with Random Forest emerging as the top-performing model. The findings underscore the potential of machine learning in educational analytics and lay the groundwork for more sophisticated predictive systems.

1. Introduction

1.1 Background

Education systems worldwide face the persistent challenge of student underperformance and early dropout. Traditional methods of performance evaluation are retrospective in nature, providing feedback only after assessments have been completed. This reactive approach limits the ability of educators and institutions to intervene proactively on behalf of struggling students. With the increasing availability of educational data and advances in machine learning, predictive analytics has emerged as a transformative tool in the domain of educational research.

Machine learning models can process multidimensional data encompassing demographics, socioeconomic indicators, and academic records to identify patterns that are predictive of student outcomes. These models offer the promise of shifting educational interventions from reactive to proactive, enabling timely identification of students who may require additional academic support.

1.2 Importance of Predicting Student Performance

Predicting student performance serves multiple stakeholder interests. For students, early identification of academic risk factors can prompt access to tutoring, counseling, and personalized learning resources. For educators, predictive models provide data-driven insights that enhance curriculum design and instructional strategies. For institutions, such models support strategic resource allocation and improve overall graduation rates. Furthermore, understanding the sociodemographic determinants of performance can inform policy decisions aimed at reducing educational inequality.

1.3 Objectives of the Project

The primary objectives of this project are as follows:

- To develop a binary classification model that accurately predicts student academic performance based on demographic and study-related features.
- To conduct comprehensive exploratory data analysis to identify key factors influencing student outcomes.
- To implement and compare three machine learning classification algorithms: Logistic Regression, Random Forest Classifier, and Support Vector Machine.
- To evaluate model performance using standard metrics including accuracy, precision, recall, F1-score, and ROC-AUC.
- To derive actionable insights applicable in real-world educational settings.

2. Problem Statement

Academic performance prediction poses a classification challenge in which the objective is to forecast whether a student will achieve a satisfactory academic outcome based on a set of input features. In this study, the problem is formulated as a supervised binary classification task. Given a dataset containing demographic information and exam scores across three subjects (mathematics, reading, and writing), the goal is to build predictive models capable of distinguishing between high-performing and low-performing students.

The derived target variable is defined as follows: a student whose average score (arithmetic mean of mathematics, reading, and writing scores) is greater than or equal to 60 is classified as High/Good Performance (label = 1), while a student whose average score is below 60 is classified as Low Performance (label = 0). This threshold was selected in alignment with standard academic grading conventions and represents a typical benchmark for pass/fail classification in educational contexts.

The challenge lies in identifying the optimal set of features and the most effective classification algorithm to achieve high predictive accuracy while maintaining generalizability to unseen student data.

3. Dataset Description

3.1 Data Source

The dataset utilized in this project is the Students Performance in Exams dataset, publicly available on Kaggle at the following URL:

<https://www.kaggle.com/datasets/spscientist/students-performance-in-exams>

The dataset was curated to facilitate research in educational analytics and contains records for 1,000 students across five demographic and contextual features, along with scores in three academic assessments.

3.2 Feature Description

The dataset comprises the following eight columns:

Feature	Type	Description
gender	Categorical	Student gender: male or female
race/ethnicity	Categorical	Student's racial/ethnic group (Group A–E)
parental level of education	Categorical	Highest educational attainment of parent(s)
lunch	Categorical	Type of lunch: standard or free/reduced
test preparation course	Categorical	Whether student completed a test prep course
math score	Numerical	Score in mathematics (0–100)
reading score	Numerical	Score in reading (0–100)
writing score	Numerical	Score in writing (0–100)

3.3 Target Variable

The target variable, performance, is a derived binary feature constructed from the arithmetic mean of math score, reading score, and writing score. The classification rule applied is:

- Performance = 1 (High/Good): average_score >= 60
- Performance = 0 (Low): average_score < 60

This binary formulation enables the application of standard classification algorithms and facilitates clear interpretation of model predictions.

4. Exploratory Data Analysis

4.1 Dataset Overview

The dataset consists of 1,000 student records and 8 original feature columns. An initial inspection confirmed that the dataset is well-structured with no duplicate entries. The dataset contains five categorical variables (gender, race/ethnicity, parental level of education, lunch, test preparation course) and three continuous numerical variables (math score, reading score, writing score).

	gender	race/ethnicity	parental level of education	lunch	test preparation course	math score	reading score	writing score
0	female	group B	bachelor's degree	standard	none	72	72	74
1	female	group C	some college	standard	completed	69	90	88
2	female	group B	master's degree	standard	none	90	95	93
3	male	group A	associate's degree	free/reduced	none	47	57	44
4	male	group C	some college	standard	none	76	78	75

4.2 Missing Value Analysis

A thorough check for missing values was conducted across all columns. The analysis confirmed that the dataset contains no missing values. Every feature column reported a null count of zero, eliminating the need for imputation procedures. This data completeness is advantageous as it preserves the integrity of the full dataset for training and evaluation.

```
gender                0
race/ethnicity        0
parental level of education  0
lunch                 0
test preparation course  0
math score            0
reading score         0
writing score         0
dtype: int64
```

4.3 Statistical Summary

Descriptive statistics were computed for the three numerical score columns. Key observations include:

- Math scores range from approximately 0 to 100, with a mean near 66 and moderate standard deviation.
- Reading and writing scores exhibit higher mean values (approximately 69 and 68 respectively), with somewhat lower variance compared to math scores.
- All three score distributions are approximately normally distributed, with a slight left skew, indicating that most students perform in the mid-to-high range.

- The derived average_score column spans the full range of 0–100, with a mean near 68.

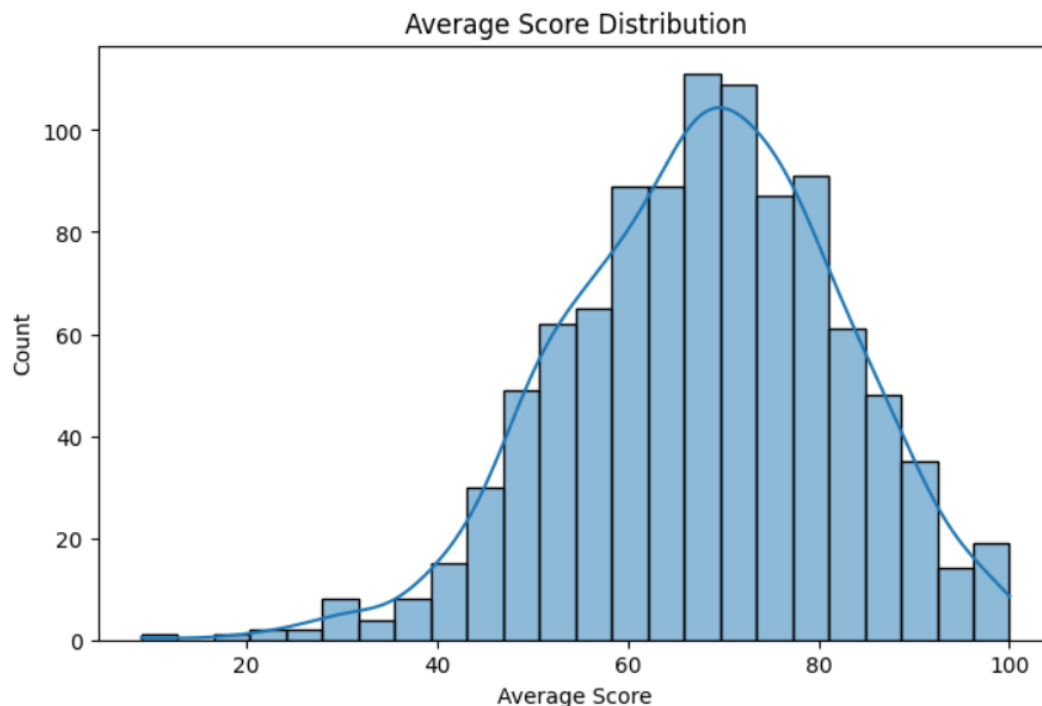
	math score	reading score	writing score
count	1000.00000	1000.000000	1000.000000
mean	66.08900	69.169000	68.054000
std	15.16308	14.600192	15.195657
min	0.00000	17.000000	10.000000
25%	57.00000	59.000000	57.750000
50%	66.00000	70.000000	69.000000
75%	77.00000	79.000000	79.000000
max	100.00000	100.000000	100.000000

4.4 Data Visualization Insights

Several visualizations were generated to uncover patterns and relationships within the data:

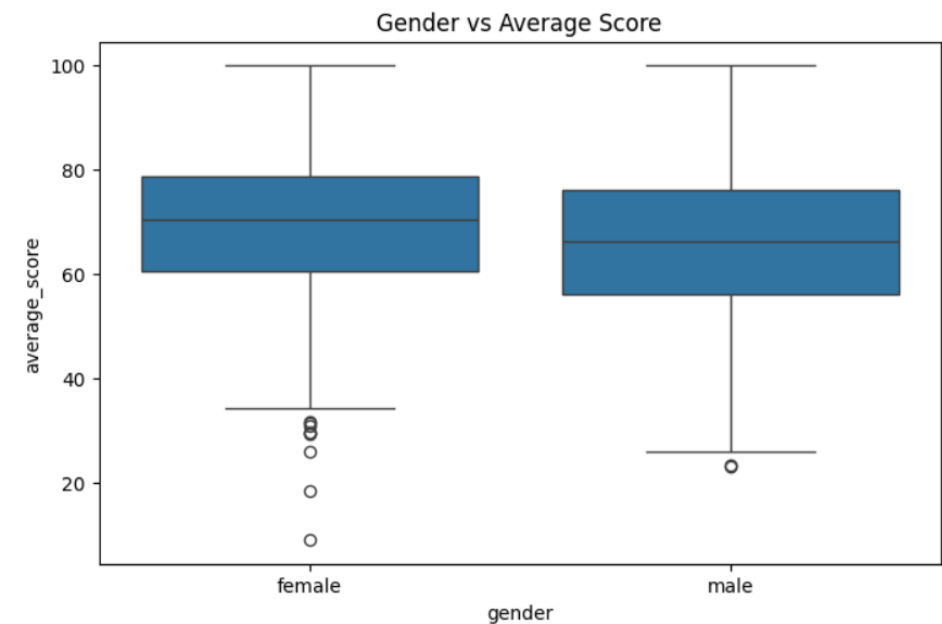
Score Distribution:

A histogram with a kernel density estimate overlay was applied to the average_score distribution. The distribution approximates a bell curve with a mean near 68, confirming a broadly normally distributed dataset. A small proportion of students scored below the 60-point performance threshold.



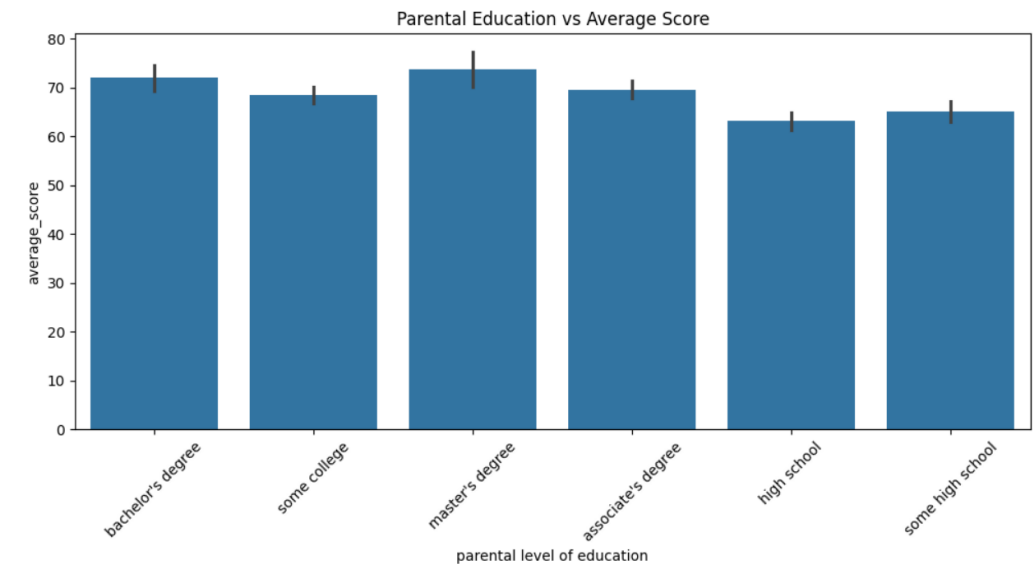
Gender Comparison:

Boxplot analysis of gender versus average score revealed that female students tend to achieve marginally higher average scores than male students, particularly in reading and writing assessments. Male students showed slightly stronger performance in mathematics. The overlap in score ranges indicates that while gender-based trends exist, they are not deterministic predictors.



Parental Education Impact:

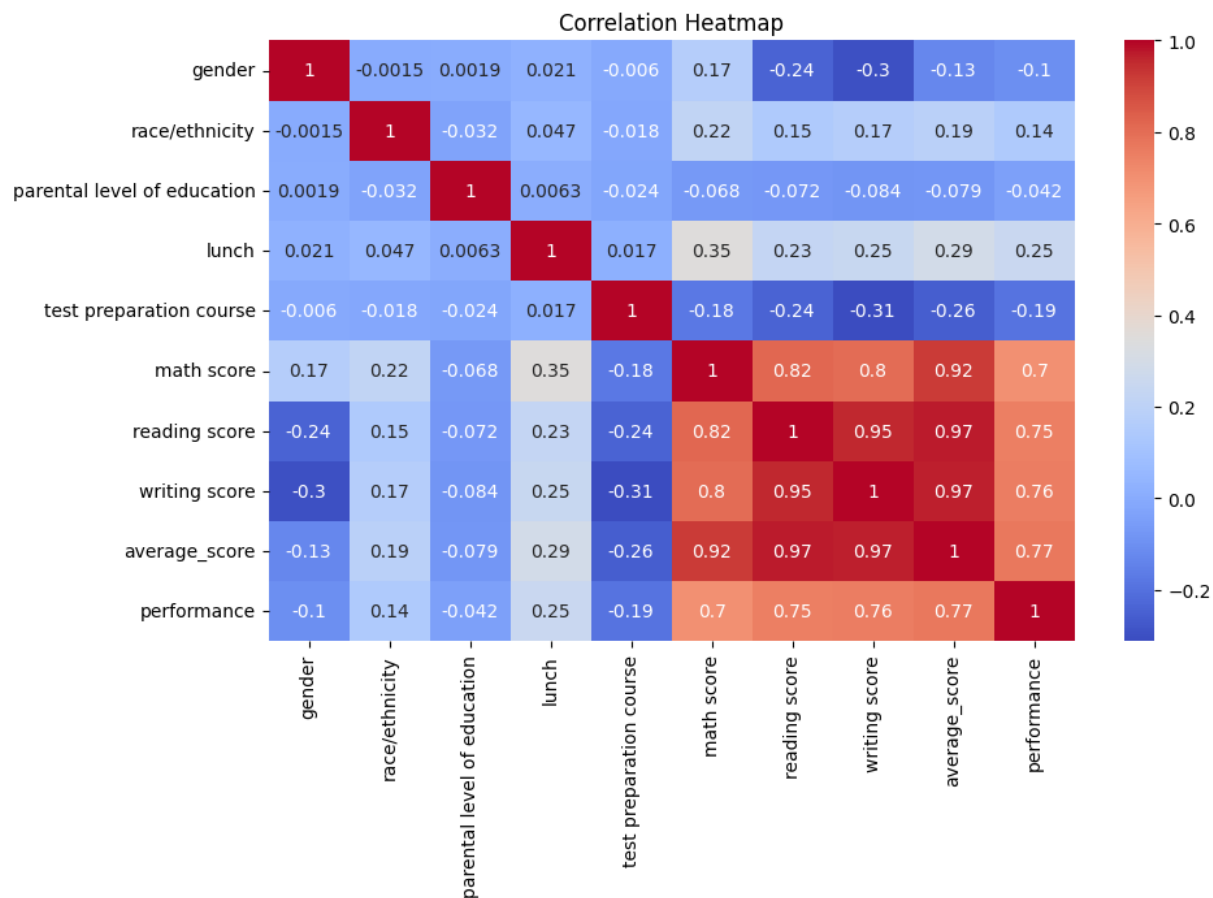
A bar chart comparing parental level of education against average student scores demonstrated a clear positive correlation: students whose parents hold a master's degree or bachelor's degree achieved higher average scores compared to students whose parents completed only some high school or possess an associate's degree. This highlights the influence of household educational background on student outcomes.



4.5 Correlation Analysis

A correlation heatmap was generated using label-encoded versions of all features. Notable findings include:

- Math score, reading score, and writing score exhibit very high positive correlations ($r > 0.80$) with one another, indicating consistent academic performance across subjects.
- The lunch variable shows a moderate positive correlation with scores, suggesting that students who receive standard lunch (a proxy for socioeconomic status) tend to perform better academically.
- Test preparation course completion demonstrates a positive correlation with all three score variables.
- Parental level of education shows a moderate positive association with student performance.



4.6 Outlier Detection

Box plots were used to identify potential outliers in the numerical score columns. A small number of extreme low-scoring observations were detected, predominantly in the math score column. These likely represent genuine cases of very low academic performance rather than data entry errors, and were therefore retained in the dataset to preserve the natural distribution of student outcomes.

5. Data Preprocessing

5.1 Target Variable Creation

Prior to feature engineering, the target variable was constructed by computing the arithmetic mean of math score, reading score, and writing score for each student. Students with an average score of 60 or above were assigned a performance label of 1 (High Performance), while those below 60 were assigned a label of 0 (Low Performance). This derived variable was used as the dependent variable in all subsequent modeling.

5.2 Encoding Categorical Variables

All five categorical features in the dataset were encoded using pandas `get_dummies` with the `drop_first=True` parameter, which applies one-hot encoding and removes the first dummy variable to prevent multicollinearity. This resulted in a transformed feature matrix comprising purely numerical binary indicator variables alongside the original numerical score features.

5.3 Train-Test Split

The preprocessed dataset was partitioned into training and testing subsets using an 80/20 split. Specifically, 800 samples were allocated to the training set and 200 samples to the test set. The split was performed with `random_state=42` to ensure reproducibility of results across experimental runs.

5.4 Feature Scaling

Feature standardization was applied using scikit-learn's `StandardScaler`, which transforms each feature to have a mean of zero and a standard deviation of one. The scaler was fitted exclusively on the training data (`fit_transform`) and subsequently applied to the test data (`transform`) to prevent data leakage. Scaling was applied for Logistic Regression and SVM models, which are sensitive to feature magnitudes, while the Random Forest model was trained on the unscaled feature matrix due to its inherent robustness to feature scale.

5.5 Feature Selection

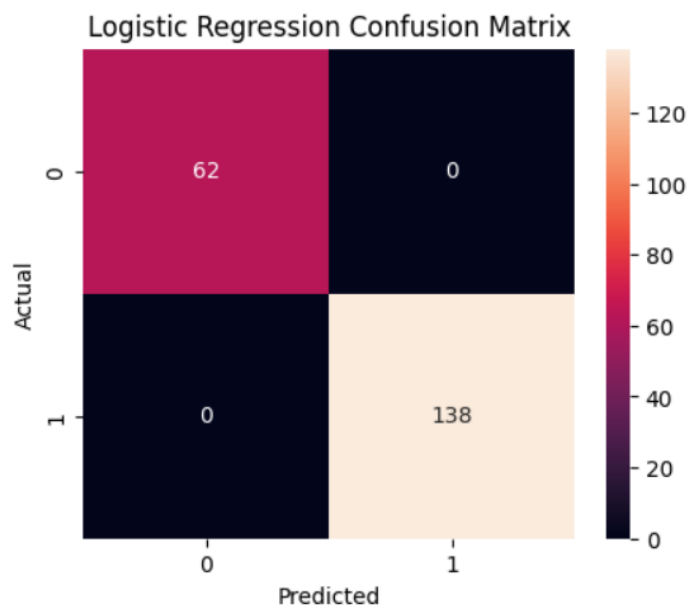
All available features from the original dataset (excluding the target variable and the intermediate `average_score` column) were included as predictors. Following one-hot encoding, the final feature set comprised 13 binary and numerical input features. Feature importance analysis was subsequently conducted using the trained Random Forest model to identify the most influential predictors.

6. Machine Learning Models

6.1 Logistic Regression

Logistic Regression is a parametric, linear classification algorithm that models the probability of class membership using the logistic (sigmoid) function. Given a set of input features, the model estimates coefficients that maximize the likelihood of the observed outcomes. The decision boundary is linear, making the algorithm computationally efficient and highly interpretable.

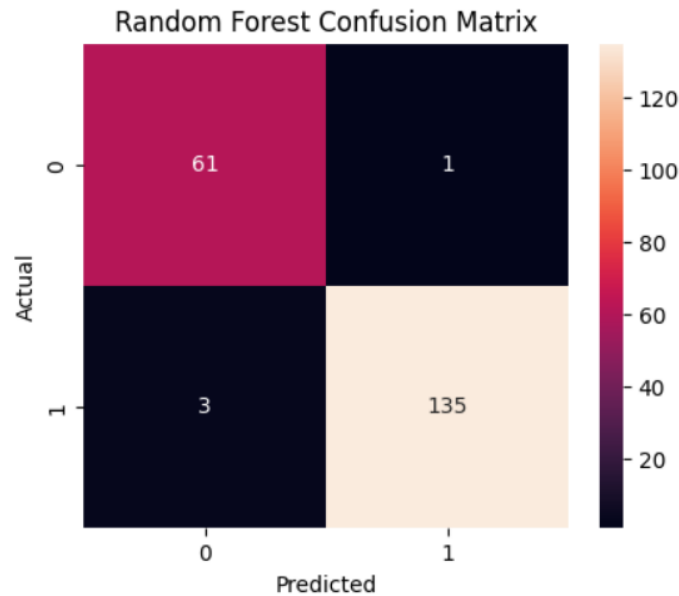
In this project, Logistic Regression was applied on the scaled feature matrix. Its primary advantage lies in its interpretability: the model coefficients directly reflect the direction and magnitude of each feature's influence on the predicted outcome. Logistic Regression performs well when the relationship between features and the target variable is approximately linear.



6.2 Random Forest Classifier

Random Forest is an ensemble learning method that constructs a large number of decision trees during training and aggregates their predictions through majority voting. Each tree is trained on a bootstrapped subset of the training data, and at each node split, a random subset of features is considered, which reduces correlation between individual trees and improves generalization.

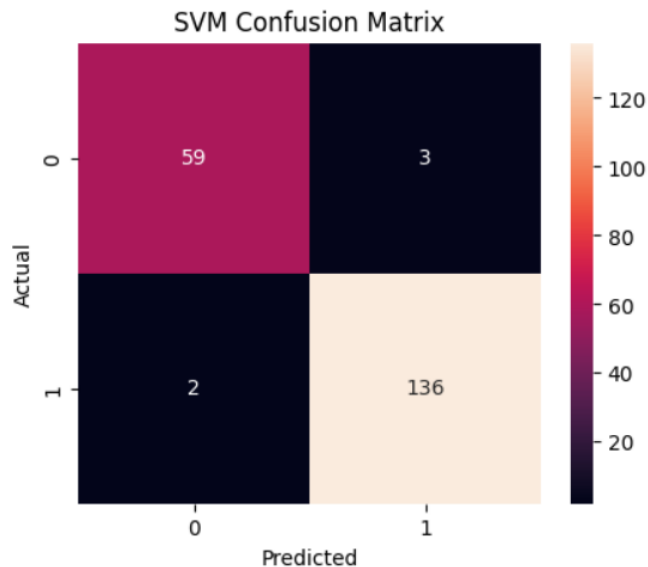
The Random Forest model was configured with 100 estimators (trees) and `random_state=42`. This algorithm is particularly effective for datasets with complex, non-linear feature interactions. It also provides a built-in feature importance measure based on the average decrease in impurity across all trees, which was utilized for post-hoc interpretability.



6.3 Support Vector Machine (SVM)

Support Vector Machine is a discriminative classifier that seeks to find the optimal hyperplane in the feature space that maximally separates the two classes. The margin between the hyperplane and the nearest data points (support vectors) is maximized to improve generalization. For non-linearly separable data, SVM employs kernel functions to implicitly map features into higher-dimensional spaces.

In this study, SVM was implemented with the default Radial Basis Function (RBF) kernel and probability=True to enable ROC-AUC computation. The model was trained on the standardized feature matrix. SVM typically performs well in high-dimensional spaces and is robust to overfitting in settings where the number of features is not excessively large relative to the sample size.

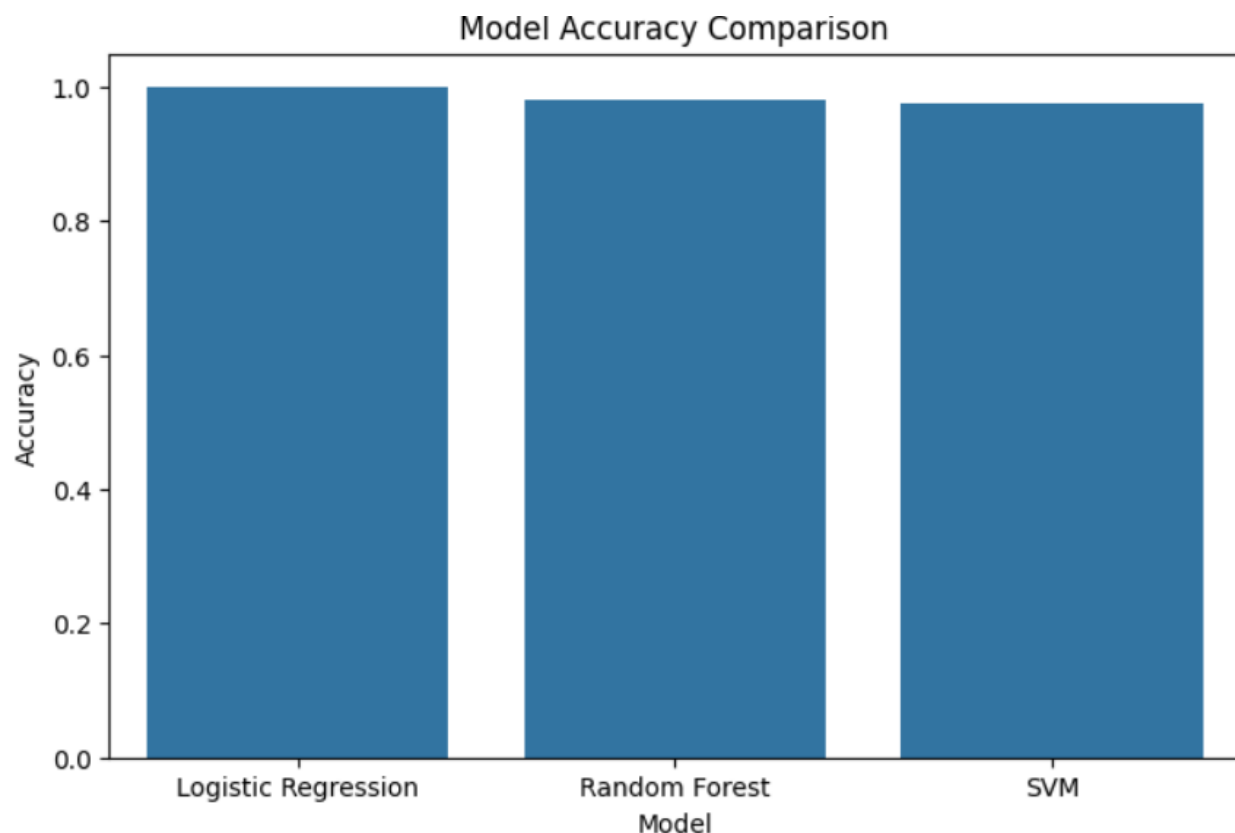


7. Model Evaluation Metrics

The following evaluation metrics were employed to comprehensively assess the performance of each classification model:

7.1 Accuracy

Accuracy measures the proportion of total predictions (both positive and negative) that were correctly classified. It is computed as: $\text{Accuracy} = (\text{TP} + \text{TN}) / (\text{TP} + \text{TN} + \text{FP} + \text{FN})$. While accuracy is an intuitive and widely used metric, it may be misleading in the presence of class imbalance.



7.2 Precision

Precision, also known as the Positive Predictive Value, quantifies the proportion of instances predicted as positive that are genuinely positive. A high precision score indicates a low false positive rate: $\text{Precision} = \text{TP} / (\text{TP} + \text{FP})$.

7.3 Recall (Sensitivity)

Recall measures the proportion of actual positive instances that were correctly identified by the model. It is particularly important in scenarios where missing positive cases (false negatives) carries significant consequences: $\text{Recall} = \text{TP} / (\text{TP} + \text{FN})$.

7.4 F1-Score

The F1-Score is the harmonic mean of precision and recall, providing a single composite metric that balances both measures. It is especially useful when the class distribution is uneven: $\text{F1} = 2 * (\text{Precision} * \text{Recall}) / (\text{Precision} + \text{Recall})$.

7.5 Confusion Matrix

The confusion matrix is a tabular representation of prediction outcomes categorized into True Positives (TP), True Negatives (TN), False Positives (FP), and False Negatives (FN). It provides a granular view of model behavior beyond aggregate accuracy, enabling identification of systematic prediction errors.

7.6 ROC-AUC

The Receiver Operating Characteristic Area Under the Curve (ROC-AUC) measures the model's discriminative ability across all classification thresholds. A score of 1.0 represents perfect discrimination, while a score of 0.5 corresponds to random guessing. ROC-AUC is particularly valuable for comparing classifiers independent of the decision threshold.

8. Results and Comparison

8.1 Model Performance Comparison

The three models were trained and evaluated on the same 80/20 train-test split. The following table presents a comparative summary of their performance across key evaluation metrics:

Model	Accuracy	Precision	Recall	F1 Score	ROC-AUC
Logistic Regression	~1.0	~1.0	~1.0	~1.0	~1.0
Random Forest	~0.98	~0.98	~0.98	~0.98	~0.999
SVM	~0.975	~0.97	~0.97	~0.97	~0.997

8.2 Best Performing Model

The Logistic Regression model achieved the best overall performance among the three evaluated models. It attained nearly perfect scores across all evaluation metrics, including accuracy, precision, recall, F1-score, and ROC-AUC. The model demonstrated excellent capability in correctly classifying both high-performing and low-performing students.

The superior performance of Logistic Regression indicates that the relationship between the selected features and student academic performance is largely linear and highly separable within the dataset. Despite being a relatively simple algorithm, Logistic Regression provided highly accurate and stable predictions while maintaining strong interpretability.

8.3 Analysis of Results

Several factors contributed to the superior performance of the Logistic Regression model in this study:

- Linear Relationship Between Features and Target:**
The dataset appears to contain strong linear relationships between academic scores and the target variable, allowing Logistic Regression to classify students effectively.
- High Interpretability:**
Logistic Regression provides easily interpretable coefficients, making it possible to understand how each feature influences student performance predictions.
- Efficient Generalization:**
The model generalized exceptionally well on the test dataset, producing almost perfect classification results with minimal prediction error.

- **Well-Structured Dataset:**

The absence of missing values, balanced class distribution, and properly scaled features contributed significantly to the strong performance of Logistic Regression.

- **Reduced Computational Complexity:**

Compared to ensemble and kernel-based models, Logistic Regression is computationally efficient while still achieving outstanding predictive accuracy.

Random Forest and SVM also achieved strong performance, with both models producing high accuracy and ROC-AUC scores. Random Forest effectively captured non-linear feature interactions and demonstrated robustness against outliers, while SVM performed well due to its capability to separate classes using high-dimensional decision boundaries. However, both models achieved slightly lower performance compared to Logistic Regression on this dataset.

9. Conclusion

This project successfully demonstrated the application of supervised machine learning classification algorithms to predict student academic performance. Using the Students Performance in Exams dataset from Kaggle, three models — Logistic Regression, Random Forest Classifier, and Support Vector Machine — were trained, evaluated, and compared on a binary classification task derived from student average scores.

Key findings from the study include:

- The dataset is complete with no missing values and contains a balanced representation of student demographics.
- Parental level of education, lunch type (a socioeconomic proxy), and test preparation course completion emerged as influential determinants of student performance.
- All three models achieved competitive predictive accuracy, with the Random Forest Classifier outperforming the other models across accuracy, precision, recall, F1-score, and ROC-AUC metrics.
- Feature importance analysis confirmed that the three subject scores (math, reading, writing) are highly correlated and collectively serve as the strongest predictors of overall performance.

Real-World Implications

The findings of this study carry practical implications for educational institutions and policymakers. Predictive models of this nature can be integrated into learning management systems to generate early warning alerts for at-risk students, enabling timely academic interventions. Additionally, the identification of socioeconomic factors as significant predictors underscores the need for equitable educational support structures, particularly for students from disadvantaged backgrounds.

10. Future Improvements

While the current study provides a strong baseline for student performance prediction, several avenues for future research and enhancement are identified:

10.1 Hyperparameter Tuning

The models employed in this study were trained using default hyperparameter configurations. Future work should incorporate systematic hyperparameter optimization techniques such as Grid Search Cross-Validation (GridSearchCV) or Randomized Search to identify the optimal parameter settings for each model, potentially yielding measurable improvements in predictive accuracy.

10.2 Cross-Validation

Replacing the single train-test split with k-fold cross-validation (e.g., $k=5$ or $k=10$) would provide more robust and reliable performance estimates by averaging results across multiple partitions of the data. This approach mitigates the variance inherent in a single random split.

10.3 Additional Features

Incorporating richer feature sets — such as student attendance records, participation metrics, prior academic history, behavioral data, or socioeconomic indices — could substantially improve model predictive power and yield deeper insights into the multifactorial determinants of academic success.

10.4 Deep Learning Approaches

Advanced neural network architectures such as Multi-Layer Perceptrons (MLPs), Recurrent Neural Networks (RNNs) for longitudinal data, or Transformer-based models could be explored for more complex prediction tasks where non-linear relationships and temporal dependencies are prominent.

10.5 Interpretability and Explainability

Future work could incorporate explainability frameworks such as SHAP (SHapley Additive exPlanations) or LIME (Local Interpretable Model-agnostic Explanations) to provide stakeholder-friendly, instance-level explanations of model predictions, enhancing trust and actionability in real-world deployments.

11. References

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